ARTIFICIAL INTELLIGENCE

A Semantic Score Generation Method for Health Misinformation Detection

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Title: A Semantic Score Generation Method for Health Misinformation Detection

1. Dual Knowledge Base Architecture

Shifted from a single Semantic Health Knowledge Graph (SHKG) to a Dual-KB system. By separating true and false claims, the system now compares evidence from both perspectives and calculates contradiction-aware features, drastically improving discriminative power.

2. Advanced Feature Engineering

Expanded from basic graph/semantic similarity metrics to a robust **12-dimensional hybrid feature set**. This includes semantic similarity, PageRank-based importance, logic margin scores, knowledge-base differences, and graph structural properties.

3. Class Imbalance Handling

While the midterm ignored class imbalance, the final version successfully addresses it by incorporating a hybrid resampling strategy using **SVM-SMOTE** and **Tomek Links**, significantly improving learning quality on imbalanced datasets.

4. Rigorous Experimental Design & Validation

Methodological flaws were corrected to ensure scientific reliability:

- Knowledge graphs are now strictly constructed using only training data, eliminating data leakage.
- Synthetic convergence demonstrations were replaced with real PageRank measurements.
- Evaluation examples were properly moved to held-out test data.

5. Enhanced Visualization & Explainable AI (XAI)

Introduced an explainability module to analyze feature contributions. The final project adds comprehensive visual analyses, including feature importance, SVM decision boundaries, PCA-based feature spaces, and real PageRank convergence, improving model interpretability.

6. Comprehensive Performance Evaluation

Evaluation was broadened beyond basic classification metrics to include balanced vs. imbalanced dataset experiments, ablation studies, and deep comparative analyses between the baseline and the proposed novelty methods.